CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which pair of materials would make the best flask lid? 1

(a) iron and copper **(b)** glass and steel **(c)** silver and gold **(d)** cork and plastic

2. A metal bowl on a table loses heat to the cold floor by: 1

(a) conduction **(b)** convection only **(c)** evaporation **(d)** radiation only

3. Which activity illustrates only radiation? 1

(a) feeling warmth from a lit bulb in a closed glass jar
(b) stirring soup
(c) boiling water
(d) heating a pan

4. A student wants a quick and even heating of water. She should: 1

(a) not heat **(b)** heat it from above
(c) heat it from the sides only **(d)** heat it from below

5. Why does the water in the middle layers of a heated beaker move in circular paths? 1

(a) conduction **(b)** evaporation
(c) radiation **(d)** convection currents

6. A pan on a stove loses heat from its sides by: 1

(a) evaporation only **(b)** convection and radiation
(c) melting **(d)** conduction only

7. A student suspects convection is absent in a vacuum. Which statement is true? 1

(a) convection is conduction **(b)** convection is radiation
(c) convection needs no medium **(d)** convection needs a fluid

8. In a vacuum flask, which transfer is still partly possible without special silvering? 1

(a) convection **(b)** evaporation **(c)** radiation **(d)** conduction

9. **Assertion (A):** Dark-coloured clothes are preferred in summer. 1
Reason (R): Dark-coloured surfaces absorb more heat radiation than light-coloured surfaces.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A thermos flask keeps hot liquids hot for a long time. 1
Reason (R): The vacuum between its double walls prevents heat transfer by conduction and convection.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does smoke from a cigarette rise in still air? 2

12. Why does the tin roof of a hut become very hot in sunlight? 2

13. Why is the roof painted white in many hot-climate houses? 2

14. Why is a metal rod with a wooden handle used for a hot-wire cutter? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why do people in cold regions build igloos with snow blocks and a small entrance? 3

16. Why does water in a metal bucket remain cooler in a shed than outside in the sun? 3

17. Explain why dust particles seen in a sunbeam in a room move. 3

OR

Explain why a pressure cooker cooks faster and how the lid and the weight help. 3

18. Describe the movement of air in a room with a window open at the top and bottom on a cold day. 3

OR

Why do some animals like elephants flap their ears? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mohan carries hot tea in a thermos flask during a trip to Shimla. The flask has double walls with a vacuum in between and shiny inner surfaces. On the same day, he wears a white shirt to stay cool in the sun. 5
(i) By which mode does heat reach Mohan from the Sun?
(ii) Why is a white shirt better in summer?
(iii) Which two modes of heat transfer does the vacuum stop?
(iv) Why are the walls of the flask made shiny?

20. Describe how to demonstrate that radiation can pass through glass but conduction needs contact. 5

OR

Prepare a note on energy saving at home using ideas of heat transfer. 5

END OF QUESTION PAPER — Total: 40 Marks